Final Report

SI190C Practical Robotics

Camera Calibration, Hand-Eye Calibration, and Line Interpolation

Scope: Numerical results are reported only where supporting files are available. The robot remained unavailable after a joint-gear failure, so simulation or planning results are kept separate from physical accuracy claims.

1 Camera Intrinsic

1.1 Method and configuration

The 640 x 480 fisheye camera was modeled with the equidistant projection r = f theta. The calibration implementation uses OpenCV fisheye calibration, a 9 x 6 inner-corner checkerboard, 25 mm square size, and sub-pixel corner refinement. The saved result contains the camera matrix K, four fisheye distortion coefficients D, and the reprojection RMS.

Image size	640 x 480 pixels
Camera model	Fisheye equidistant model
Calibration target	9 x 6 inner corners; 25 mm square size
Saved RMS	0.7553 pixels

1.2 Calibration result

```
K = [ 354.569    0.000  281.249 ]
    [   0.000  351.722  200.014 ]
    [   0.000    0.000    1.000 ]
D = [ 0.17083  -0.04136   0.08014  -0.06016 ]
Reprojection RMS = 0.7553 px
```

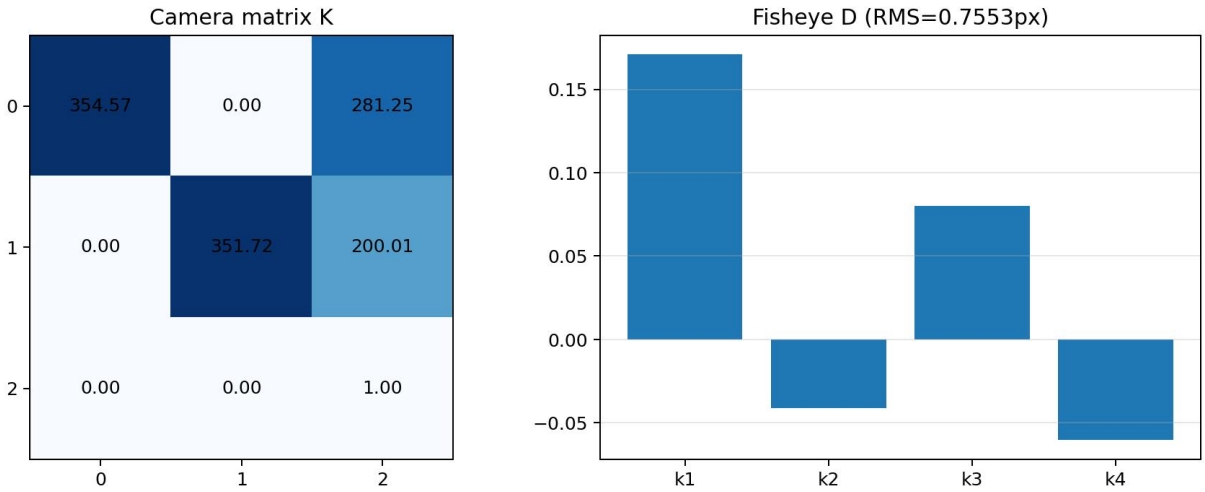


Figure 1. Camera matrix and fisheye distortion coefficients.

1.3 Accuracy analysis

The RMS of 0.7553 px is below the course acceptance threshold of 0.8 px, so the numerical calibration converged successfully. It is nevertheless close to the threshold and does not reach the commonly used excellent

level of 0.5 px. The result is suitable for the following algorithm tests, while additional views near image boundaries and improved focus would be expected to reduce peripheral fisheye error.

Reproducibility limit: The raw checkerboard images and an original-versus-undistorted image pair were not retained. Therefore, view diversity and qualitative undistortion quality cannot be independently checked from the submitted files.

2 Hand-eye Calibration

2.1 Method and data

For the eye-in-hand configuration, each record contains the end-effector pose in the robot base frame T_be and the marker pose in the camera frame T_cm. The fixed-marker relationship is T_bo = T_be T_ec T_cm. Relative motions were converted to A X = X B and solved with the Tsai method in OpenCV. The data file contains 12 records and 10 unique pose pairs.

Item	Value
Recorded pairs	12
Unique pairs	10 (two duplicate pairs)
End-effector X range	[-0.279, 0.167] m
End-effector Y range	[-0.295, 0.327] m
End-effector Z range	[0.021, 0.468] m

2.2 Calibration result

```
T_ec = [ -0.999999996  -0.000092654   0.000   0.000 ]
       [  0.000092654  -0.999999996   0.000   0.406 ]
       [  0.000000000   0.000000000   1.000   0.390 ]
       [  0.000000000   0.000000000   0.000   1.000 ]
Translation = [0.000, 0.406, 0.390] m
Rotation    = approximately 180 deg about Z
```

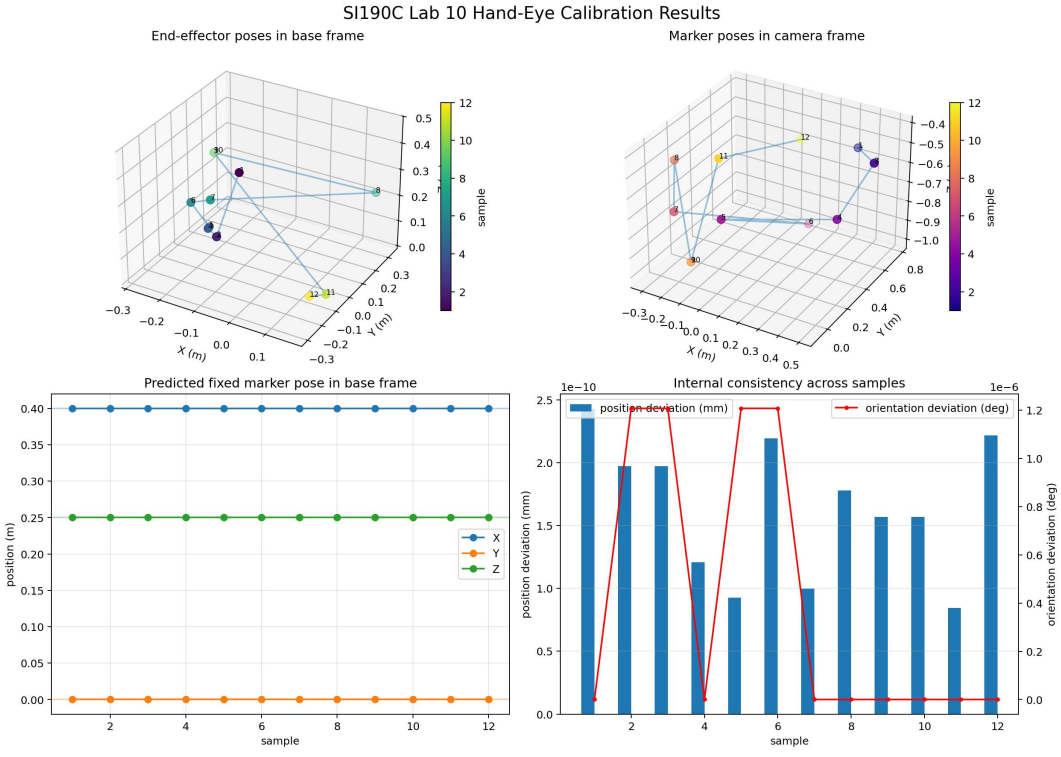


Figure 2. Pose distribution and internal consistency of the saved calibration set.

2.3 Accuracy analysis

Back-substitution gives a constant marker location of approximately [0.400, 0.000, 0.250] m in the base frame. The saved arrays produce position deviations below 2.5e-10 mm and orientation deviations below 1.3e-6 deg. These values demonstrate algebraic consistency of the transform chain.

Interpretation: The calibration arrays match the hard-coded simulator values to floating-point precision. The machine-precision residual is therefore a simulation consistency check, not independent robot accuracy. Mechanical joint gears failed and remained unavailable after two replacements; physical verification against the 5 mm and 3 deg targets could not be completed.

3 Line Interpolation

3.1 Trajectory design

The ROS 2 pose publisher defines three Cartesian waypoints and generates the closed sequence A to B to C to A. Position is interpolated with Lerp, while orientation is interpolated with normalized quaternion Slerp. For each segment, the program publishes the segment start and 20 interior samples; the next segment provides the previous endpoint. This gives 21 samples per segment and 63 samples per cycle at the configured 1 Hz publication rate.

Point	X (m)	Y (m)	Z (m)
A	0.125	-0.194	-0.049
B	0.014	-0.307	-0.030
C	0.088	-0.034	0.177

Segment	Distance (m)	Requirement
A-B	0.15953	> 0.15 m
B-C	0.35051	> 0.15 m
C-A	0.27937	> 0.15 m

3.2 Motion trajectory

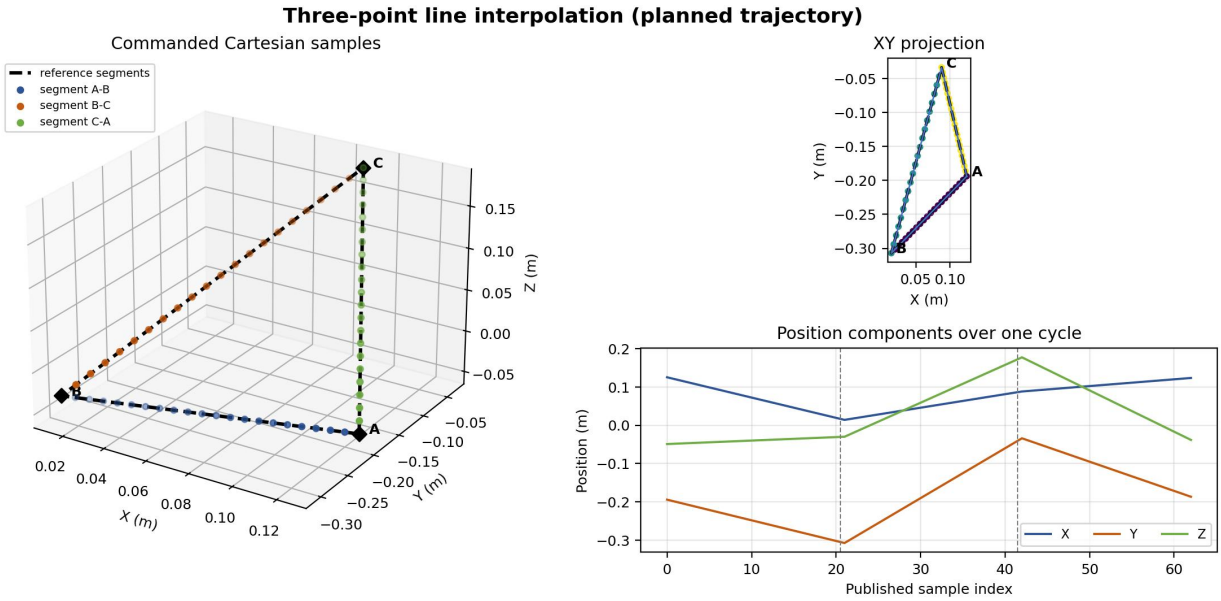


Figure 3. Planned three-segment Cartesian trajectory. Dots are commanded samples; dashed lines are references.

3.3 Accuracy analysis

Using the active segment for each sample, the maximum numerical distance from a generated point to its reference line segment is 7.972e-14 mm. This near-zero value is expected because the samples are constructed directly from the line equation; it verifies the interpolation geometry only and is not a controller-tracking measurement.

Hardware data unavailable: No executed fk_poses.csv, ROS 2 bag, or ArUco-measured trajectory is available, so physical MAE, RMSE, maximum error, repeatability, velocity, and acceleration cannot be reported. The current publisher also uses uniform time steps and different waypoint orientations. A complete Lab 11 execution should use quintic time scaling, fixed orientation, phase/cycle tags, two cycles, and active-segment error analysis after the robot is repaired.

4 Bonus: Gripper Design

CAD-level completion status

A compact two-finger gripper was completed as an optional CAD concept. It uses an MG996R-size servo, double-arm horn, two links, pivoting jaws, ribbed pads, and a 100 x 65 x 5 mm frame. The archive contains 11 SolidWorks parts, a 15-component assembly, STEP export, and previews; nominal opening is approximately 36 mm.

This fixed open-position model was not manufactured or load-tested and does not implement the specified stepper actuator. Archive: MG996R_parallel_gripper_design.zip.

